Lab 3: Querying Audit Logs with the System Table

Description

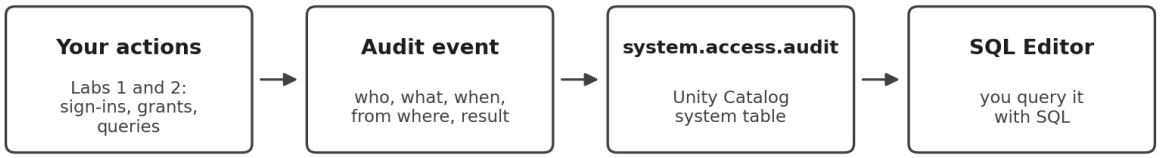
Every sign-in, grant and query you made in Labs 1 and 2 was recorded as an audit event. In this lab you find those events in the system.access.audit table using SQL, and answer simple audit questions: who signed in, who changed permissions, and who was denied access.

Learning Objectives

- Explain what the audit system table records
- Read the key columns of system.access.audit
- Find sign-ins, permission changes and denied requests with SQL
- Tell workspace-level events from account-level events

Step by Step Guide

How it works



Covers every workspace plus account-level events. Kept for 365 days. Events can take up to an hour to appear.

Your lab values

Item	Value
Prefix	Same as Labs 1 and 2 (for example bob)
Your catalog	<prefix>_catalog
Test users	<prefix>-engineer1, <prefix>-analyst1
Compute	Shared serverless SQL warehouse, in SQL Editor

Tip: Run each query on its own in SQL Editor. Replace <prefix> before running. All queries look at the last 30 days; increase the number if Labs 1 and 2 were earlier.

Step 1: Confirm you can read the audit table

1. Open **Catalog** and expand system > access > audit. Open the **Columns** tab.
2. In SQL Editor, run:

```
SELECT count(*) AS events_last_7_days
FROM system.access.audit
WHERE event_date >= current_date() - 7;
```

> **Check:** The query returns a number greater than zero.

Step 2: Look at the key columns

```
SELECT event_time, workspace_id, service_name, action_name,
       user_identity.email, source_ip_address,
       request_params, response.status_code
FROM system.access.audit
WHERE user_identity.email = current_user()
  AND event_date >= current_date() - 30
ORDER BY event_time DESC
LIMIT 20;
```

Column	What it tells you
event_time	When it happened (UTC)
service_name / action_name	Which service and what action, for example unityCatalog / updatePermissions
user_identity.email	Who did it
source_ip_address	Where the request came from
request_params	Details of the request, such as the object name
response.status_code	200 means success; anything else means it failed or was denied
workspace_id	0 means an account-level event

> **Check:** current_user() returns your email, so every row shows an action you performed.

Step 3: Summarise everything you did

```
SELECT service_name, action_name, count(*) AS events
FROM system.access.audit
WHERE user_identity.email = current_user()
  AND event_date >= current_date() - 30
GROUP BY ALL
ORDER BY events DESC;
```

> Check: You see services such as unityCatalog, accounts and databrickssql.

Step 4: Who signed in as your test users? (Lab 1)

```
SELECT event_time, user_identity.email, action_name,
       source_ip_address, response.status_code
FROM system.access.audit
WHERE service_name = 'accounts'
  AND action_name IN ('aadBrowserLogin', 'login', 'aadTokenLogin', 'tokenLogin')
  AND (user_identity.email LIKE '<prefix>-%'
       OR request_params.user LIKE '<prefix>-%')
  AND event_date >= current_date() - 30
ORDER BY event_time DESC;
```

> Check: You see sign-ins for engineer1 and analyst1 from Labs 1 and 2, with the IP address of your machine.

Step 5: Find your account console actions (Lab 1)

Adding groups and assigning them to the workspace happened in the account console, so these events have workspace_id 0.

```
SELECT event_time, service_name, action_name, request_params
FROM system.access.audit
WHERE workspace_id = 0
  AND service_name <> 'unityCatalog'
  AND user_identity.email = current_user()
  AND event_date >= current_date() - 30
ORDER BY event_time DESC;
```

> Check: Look inside request_params for your group names. Remember these: in Lab 4 you will see that Log Analytics does not receive account-level events.

Step 6: Who changed permissions on your catalog? (Lab 2)

```
SELECT event_time,
       user_identity.email AS who,
       request_params.securable_type,
       request_params.securable_full_name,
       request_params.changes
FROM system.access.audit
WHERE service_name = 'unityCatalog'
  AND action_name = 'updatePermissions'
  AND request_params.securable_full_name LIKE '<prefix>_catalog%'
  AND event_date >= current_date() - 30
ORDER BY event_time;
```

> Check: Each GRANT and REVOKE from Lab 2 appears, with the group and privileges in the changes column.

Step 7: Who touched your catalog, and what did they do?

```
SELECT user_identity.email AS who, action_name, count(*) AS events
FROM system.access.audit
WHERE service_name = 'unityCatalog'
  AND to_json(request_params) LIKE '%<prefix>_catalog%'
  AND event_date >= current_date() - 30
GROUP BY ALL
ORDER BY who, events DESC;
```

> Check: You see yourself (the owner) plus engineer1 and analyst1.

Step 8: Find denied requests

```
SELECT event_time, user_identity.email AS who, action_name,
       response.status_code, response.error_message
FROM system.access.audit
WHERE service_name = 'unityCatalog'
  AND to_json(request_params) LIKE '%<prefix>_catalog%'
  AND response.status_code <> 200
  AND event_date >= current_date() - 30
ORDER BY event_time DESC;
```

> Check: You see analyst1's failed attempts from Lab 2, for example before you granted access and when they tried to insert rows or create a table.

Outcomes

- You can find the audit system table and explain its key columns

- You traced your own Lab 1 and Lab 2 activity with SQL
- You found sign-ins, permission changes and denied requests
- You know that account-level events have workspace_id 0